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B.M.S. College of Engineering, Bengaluru-560019
Autonomous Institute Affiliated to VTU

Programme: B.E., Semester: III	Duration: 3 hrs.
Branch: A.I. & Machine Learning	Max Marks: 100
Course Code: 23AM3PCPSM	Course: Probability and Statistics for ML

	Questions	CO	PO	Marks

Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.

	UNIT - I			
1	a) Distinguish between Joint probability and Marginal probability in the context of Machine Learning, explaining why understanding these concepts is crucial for accurate predictions, and provide an example calculation to illustrate this difference.	CO1	PO1	10
	b) What mathematical formulation can be used to justify the design of a probabilistic model that effectively evaluates uncertainty in machine learning systems, and how does it relate to the concept of conditional probability?	CO1	PO2	10
	OR			
2	a) Distinguish between joint and conditional probability in machine learning contexts, comparing their implications on model performance and interpretability under varying levels of uncertainty.	CO2	PO1	10
	b) What theoretical justification can be made to design and implement a machine learning model that optimizes for conditional probability, rather than marginal or joint probability, in order to effectively evaluate uncertainty in complex prediction tasks?	CO2	PO2	10
	UNIT - II			
3	a) What is the difference in application between the Bayes Theorem and Maximum a Posteriori Estimation, specifically distinguishing between the two methods when modeling complex hypotheses with multiple classes?	CO1	PO1	10
	b) Design a Naive Bayes classifier for multiclass classification problems and justify its suitability by evaluating its assumptions on conditional probability distributions through the lens of prior and conditional probabilities, while comparing it to more complex models, such as maximum a posteriori estimators or traditional discriminant analysis.	CO1	PO2	10
	OR			
4	a) What are the key differences in calculating prior and conditional probabilities for a Simplified Naive Bayes classifier versus a traditional Conditional Probability Model of Classification?	CO2	PO1	10

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	b) Design a Bayesian classifier that can be used to classify handwritten digits as either 0 or 1 using a simplified version of the Naive Bayes model; justify your choice of simplifying the features by providing a clear explanation of the assumptions made regarding prior and conditional probabilities.			
	UNIT - III			
5	a) Distinguish between the conditions under which a discrete random variable X and another discrete random variable Y are said to be independent, and give an example of two random variables that satisfy this condition.	CO1	PO1	10
	b) What is the expected value and variance of a binomial random variable with parameters n = 5 and p = 0.3, and justify your answer using the properties of discrete random variables?	CO1	PO2	10
	OR			
6	a) What is the primary difference between the Binomial and Geometric distributions when modeling the number of trials required to achieve a specified number of successes, and how do these differences impact their application in probability theory?	CO2	PO1	10
	b) Design a statistical model to evaluate the independence of two discrete random variables, X and Y, with different probability distributions, justifying your assumptions about their joint distribution using the concept of conditional entropy.	CO2 CO2	PO2 PO2	10 10
	UNIT - IV			
7	a) Distinguish between the probability density functions of the exponential distribution and the normal distribution, and explain how the exponential distribution can be viewed as a special case of the normal distribution when its mean is zero and variance is one, in order to justify why the normal distribution often serves as an approximation for the exponential distribution in certain scenarios.	CO1	PO1	10
	b) Design an experiment to compare the performance of different probability distributions in modeling real-world continuous random variables, including Pareto distribution and normal distribution, while justifying your choice of each distribution and explaining how you will			

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	evaluate their accuracy using statistical measures such as mean squared error (MSE) and skewness.			
	OR			
8	a) Distinguish between the probability density functions of the exponential and normal distributions, considering their respective parameters, and compare them to the Pareto distribution in terms of their shapes and behaviors for very large values.	CO2	PO1	10
	b) Design an experiment to evaluate the effect of the central limit theorem on a Pareto distribution with shape parameter =2 and scale parameter =1/2, and justify why you believe this choice is appropriate for assessing the underlying assumptions of the CLT.	CO2	PO2	10
	UNIT - V			
9	a) Compare the concepts of population and sample parameters in probability theory as outlined by Michael Baron in "Probability and Statistics for Computer Scientists" with the practical applications discussed by Jason Brownlee in "Probability for machine learning", distinguishing between how to accurately represent real-world data using both approaches.	CO1 CO1	PO1 PO2	10 10
	b) Design a statistical model to estimate the population mean and variance of a real-world dataset, justifying your choice of method and explaining how it would be applied in a machine learning algorithm to make predictions on unseen data.	CO1	PO2	10
	OR			
10	a) What is the primary difference in perspective between analyzing a population and analyzing a sample of data, with reference to how parameters versus statistics are defined and measured?	CO2	PO1	10
	b) Design a statistical model to predict customer churn based on demographic and transactional features, justifying the use of quantiles and standard deviation as relevant metrics in this context.	CO2	PO2	10